

Name _____ Date _____ Period _____

When Is an Inverse a Function?

Algebra II | Unit 1 | Day 12 | TEK A2.2.C

Objective: I can decide whether the inverse of a function is itself a function, and restrict the domain of the original when it is not.

Part 1 — The Problem with $f(x) = x^2$

Fill in the table, then look for repeats in the bottom row.

x	-3	-2	-1	0	1	2	3
$f(x) = x^2$							

Which outputs appear more than once? _____ So f is _____ one-to-one.

Now swap to build the inverse. Both $(1, 1)$ and $(1, -1)$ appear. Is that a function? _____

Why not? _____

Written out, the inverse relation is $y = \pm \sqrt{x}$, and the $\pm$ is exactly the problem.

Part 2 — Fix It with a Restriction

Keep only half of the parabola, so no output repeats.

Restriction on $f(x) = x^2$	Which half of the parabola	Then $f^{-1}(x) =$
$x \geq 0$	_____	_____
$x \leq 0$	_____	_____

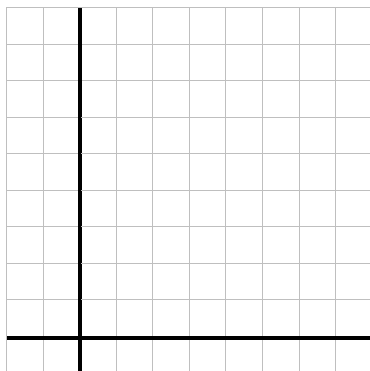
Either restriction works. The usual choice is $x \geq 0$, which gives the familiar $f^{-1}(x) = \sqrt{x}$.

Restricting the domain of f also restricts the _____ of f^{-1} .

Part 3 — Graph the Restricted Pair

Each square is 1 unit. Graph only the half of x^2 you kept, then its inverse, plus $y = x$.

$f(x) = x^2$ for $x \geq 0$, and $f^{-1}(x) = \sqrt{x}$



Record it.

Domain of restricted f : _____

Range of restricted f : _____

Domain of f^{-1} : _____

Do they mirror across $y = x$? _____

Part 4 — Which Pairs Need No Restriction?

Some parent functions are already one-to-one, so their inverses are functions with no help.

Function	One-to-one?	Restriction needed?	Its inverse
$f(x) = x^2$	_____	_____	_____
$f(x) = x^3$	_____	_____	_____
$f(x) = 2^x$	_____	_____	_____
$f(x) = 2x + 1$	_____	_____	_____
$f(x) = x $	_____	_____	_____

Pattern: functions that only _____ (or only decrease) are one-to-one. Ones that turn around, like x^2 and $|x|$, are not.

Part 5 — Quick Check

1. Is the inverse of every function also a function? _____
2. Which test tells you the inverse will be a function? _____
3. Why does x^3 need no restriction but x^2 does? _____

Practice

Decide whether a restriction is needed, and give one if so.

1. Does $f(x) = |x|$ have an inverse that is a function with no restriction? _____
2. Restrict $f(x) = |x|$ so its inverse is a function. Give the restriction and the inverse.

3. Is $f(x) = x$ to the 4th power one-to-one? _____ Give a restriction that fixes it. _____
4. True or false: $f(x) = x^3 + 1$ needs a domain restriction for its inverse to be a function.

5. Explain, using the horizontal line test, why $f(x) = x^2 - 4$ has an inverse that is not a function.

6. Give one restriction (there are two possible) that makes $f(x) = x^2 - 4$ a one-to-one function.

Exit Ticket

Consider $f(x) = x^2 - 4$ with no restriction.

Is its inverse a function? _____ Give a restriction that fixes it: _____

Explain in one sentence why the restriction is needed: _____